

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C++, TypeScript, JavaScript, SQL
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets, Raft Consensus Algorithm
- **Infrastructure & DB:** Docker, PostgreSQL, Git, Linux

TECHNICAL PROJECTS

TitanSwarm

Jan 2026 - Present

Personal Project · Python, Go, LangChain, FAISS

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot in Python, enabling the discovery and parsing of thousands of Software Engineering postings with advanced anti-bot evasion via concurrent JobSpy scrapers.
- Built a zero-hallucination RAG engine using LangChain and OpenAI APIs, delivering uniquely tailored, **ATS-optimized** PDF resumes by leveraging only verified user data.
- Developed TitanStore, a distributed memory bank in Go, implementing the Raft consensus protocol and a custom Write-Ahead Log (WAL) to guarantee application state durability and fault tolerance across network failures.

TitanStore

Jan 2026 - Present

Personal Project · Go

- Implemented the Raft consensus algorithm in Go to elect a leader, replicate writes across a cluster, and survive node failures for a distributed key-value database.
- Ensured crash-safe recovery and fault tolerance by backing the distributed database with a binary Write-Ahead Log and atomic snapshots.
- Developed a highly available, replicated storage backend for a file-sync daemon, demonstrating robust **systems** design.

SFU-Course-Tracker

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Optimized deployments and reduced infrastructure costs by migrating a full-stack platform to AWS (EC2) and implementing **Docker** container orchestration.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025